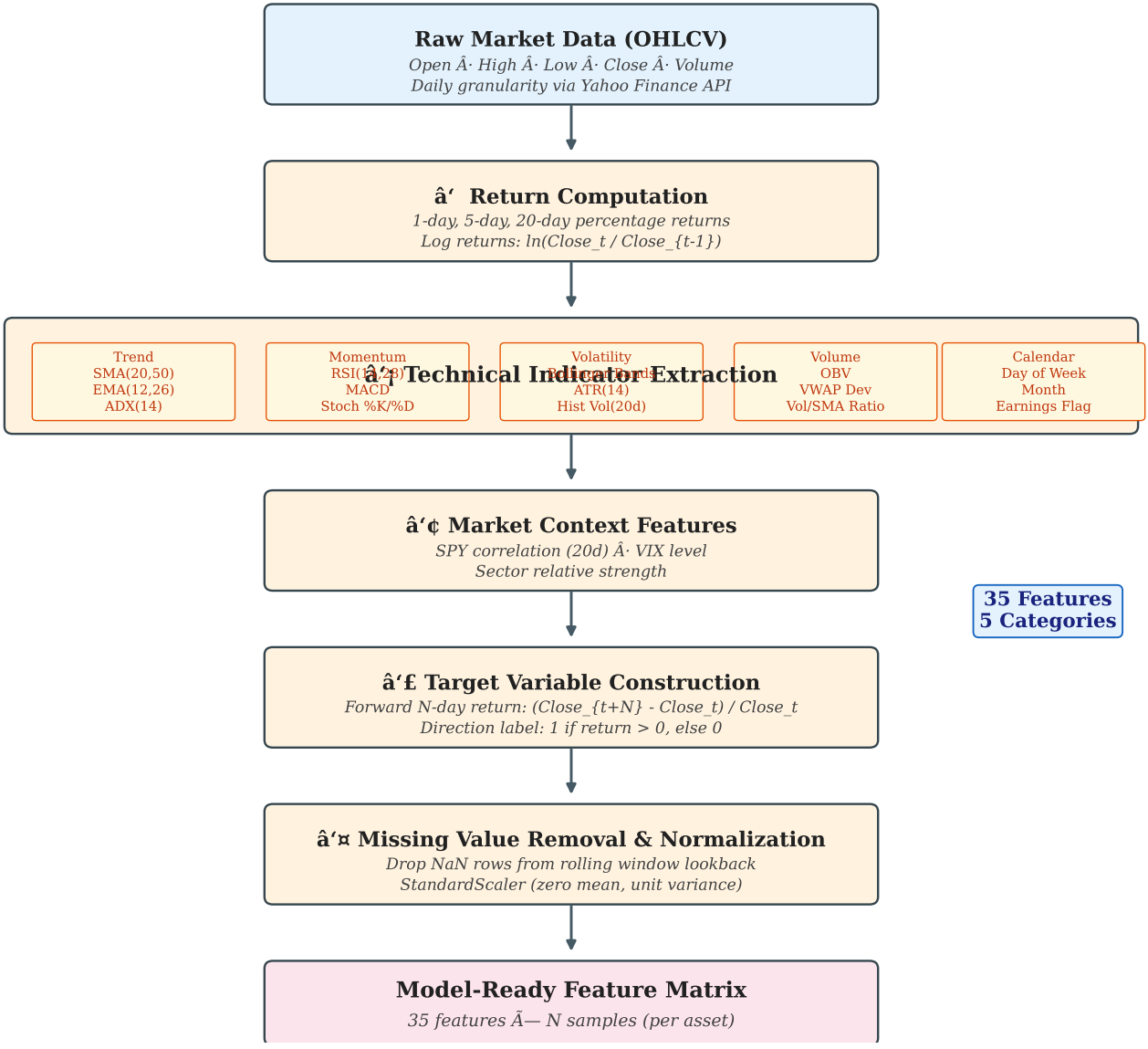

Data Preprocessing & Feature Engineering Pipeline
Raw Market Data (OHLCV)
Open Â· High Â· Low Â· Close Â· Volume
Daily granularity via Yahoo Finance API
â‘ Return Computation
1-day, 5-day, 20-day percentage returns
Log returns: ln(Close_t / Close_{t-1})
â‘¡ Technical Indicator Extraction
Trend
SMA(20,50)
EMA(12,26)
ADX(14)
Momentum
RSI(14,28)
MACD
Stoch %K/%D
Volatility
Bollinger Bands
ATR(14)
Hist Vol(20d)
Volume
OBV
VWAP Dev
Vol/SMA Ratio
Calendar
Day of Week
Month
Earnings Flag
â‘¢ Market Context Features
SPY correlation (20d) Â· VIX level
Sector relative strength
35 Features
5 Categories
â‘£ Target Variable Construction
Forward N-day return: (Close_{t+N} - Close_t) / Close_t
Direction label: 1 if return > 0, else 0
â‘¤ Missing Value Removal & Normalization
Drop NaN rows from rolling window lookback
StandardScaler (zero mean, unit variance)
Model-Ready Feature Matrix
35 features Ã— N samples (per asset)